



Lab 0

Fundamentals of Digital Logic

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Contents of this slide are provided by NTOU VLSI Lab601

Versions

- V1.2版本 109 俞彥維/楊翔鈞 研究生製作
- V1.3版本 110 俞彥維/林孟翰 研究生製作
- V1.4版本 111 林孟翰/王瀚緯/潘奕璇 研究生製作
- V1.5版本 112 潘奕璇/李佳勳 研究生製作
- V2.0版本 113 潘奕璇/李佳勳 研究生製作 (英文版)

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Lab601嚴茂旭教授實驗室提供教學及研究使用

※不提供商業用途※

Objectives

- Learn about the operations of the instruments
- Learn about properties of the passive components
- Learn about the resistor series-parallel connection method,
- Learn about the calculation of resistance in a series-parallel circuit
- Learn about the Kirchhoff voltage law and Kirchhoff current law
- Learn about the resistor color code
- Learn about properties of the active components
- Learn about the use of breadboard
- Learn about basic logic gates

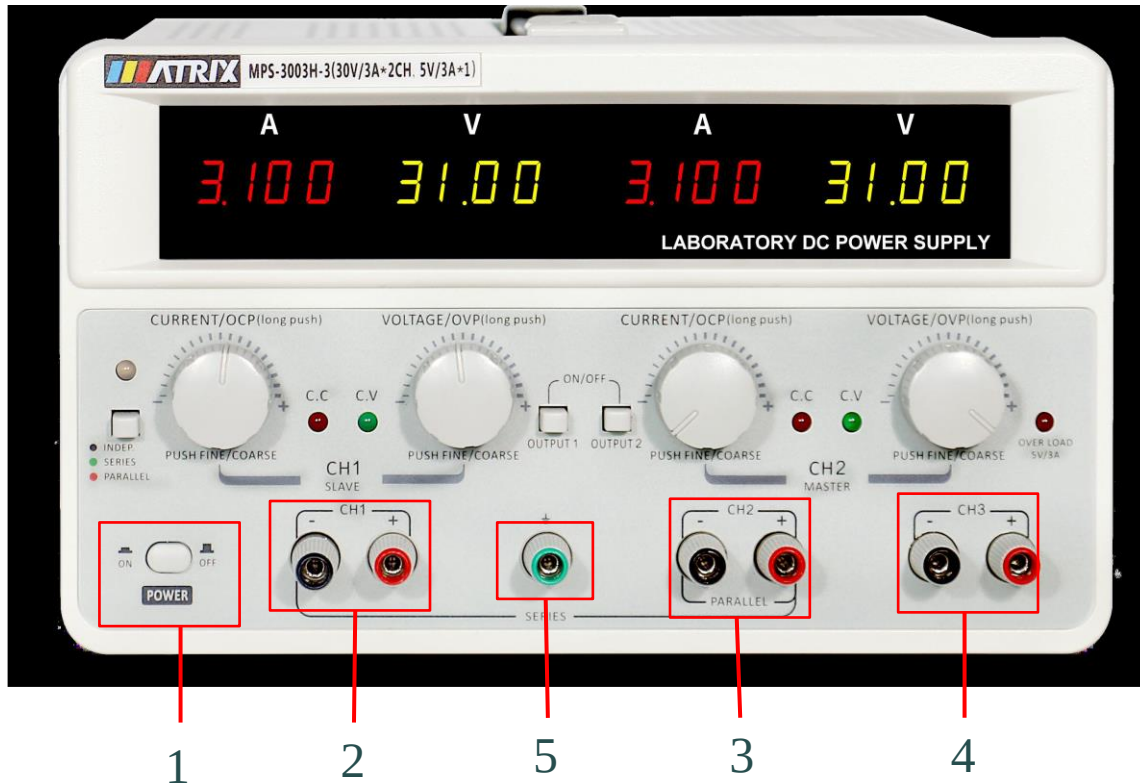
Operations of Instruments

- Power supply
- Multimeter
- Function generator
- Oscilloscope

Power Supply (1/2)

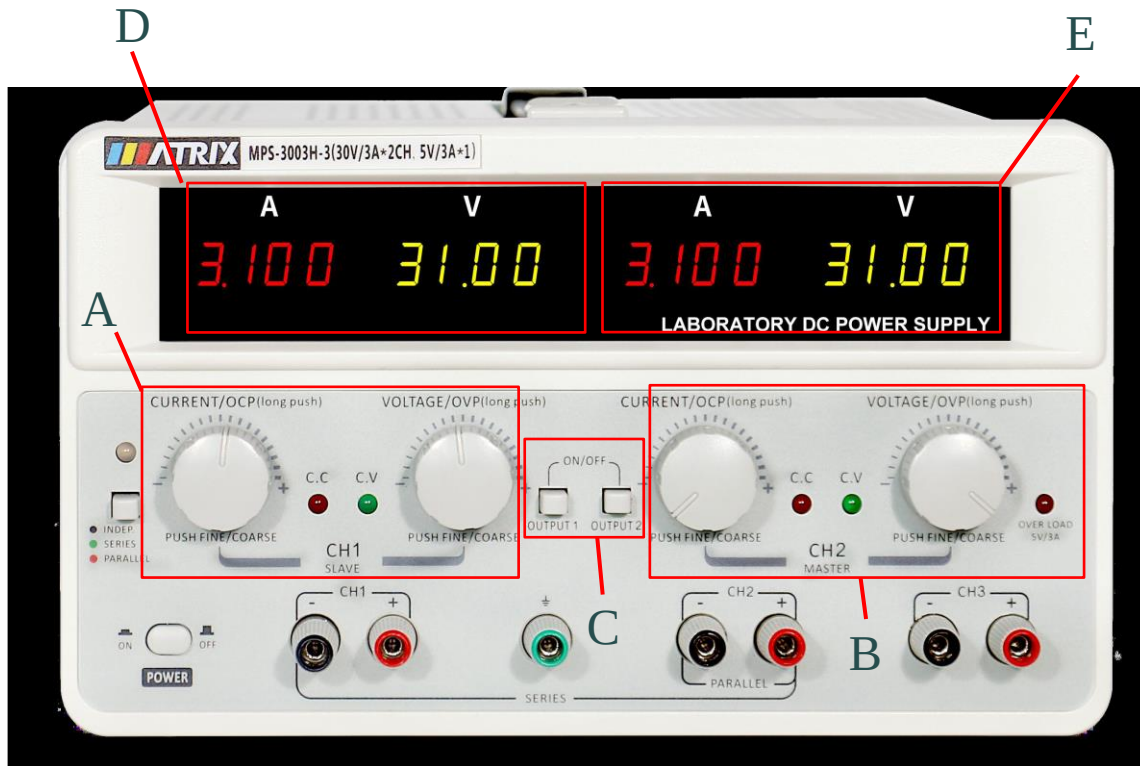
V: Volt

A: Ampere



1. On/Off switches
2. CH1 (Slave) output terminals:
 - + (red) is positive
 - – (black) is negative (or ground)
 - Voltage (0V ~ 30V) and current (0A ~ 3A)
3. CH2 (Master) output terminals:
 - + (red) is positive
 - – (black) is negative (or ground)
 - Voltage (0V ~ 30V) and current (0A ~ 3A)
4. CH3 output terminals:
 - + (red) is positive
 - – (black) is negative (or ground)
 - Fixed voltage of 5V and current 3A.
5. Ground

Power Supply (2/2)



- A. Knobs to adjust the current (left) and voltage (right) of CH1.
- B. Knobs to adjust the current (left) and voltage (right) of CH2.
- C. Output switch for CH1 (left) and CH2 (right).
- D. The display window for current and voltage (green) for CH1.
- E. The display window for current and voltage (green) for CH2.
- Knob adjustment:
 - Press the knob to select the number of digits to be adjusted
 - Rotate the knob to adjust the value
 - Save the adjustment result if you don't make it for 5 seconds

Multimeter

Voltage Measurement



- Turn the center knob to “V” position.
- The red probe should be connected to the rightmost terminal.
- The black probe should be connected to COM.

Multimeter

Resistance Measurement



- Turn the center knob to “ Ω ” position.
- The red probe should be connected to the rightmost terminal.
- The black probe should be connected to COM.

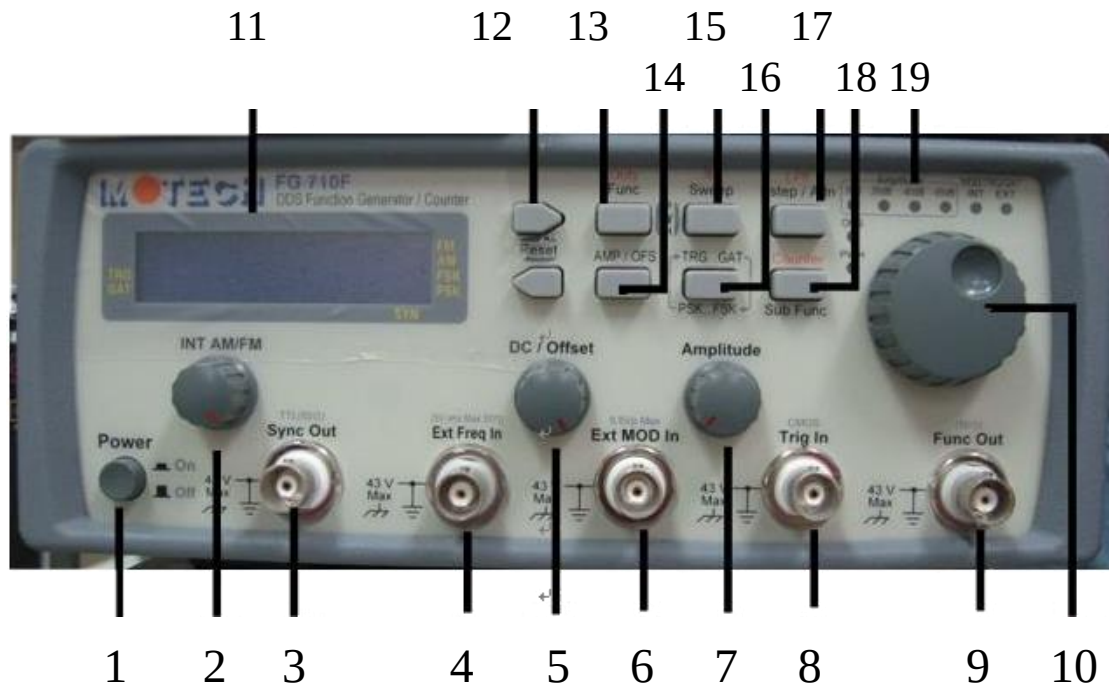
Multimeter

Current measurement



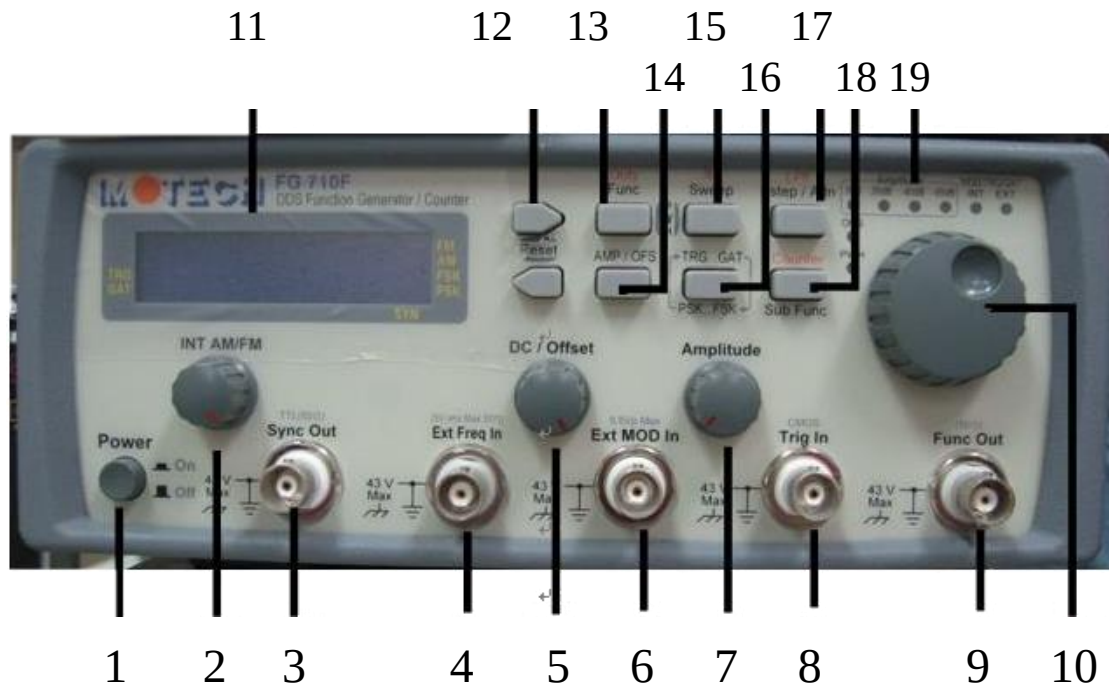
- Tune the center knob to “20A” or “mA” position.
- The red probe should be connect to the leftmost terminal, if the knob is turned to “20A”
- The black probe should still be connected to COM.

Function Generator (1/2) (Optional)



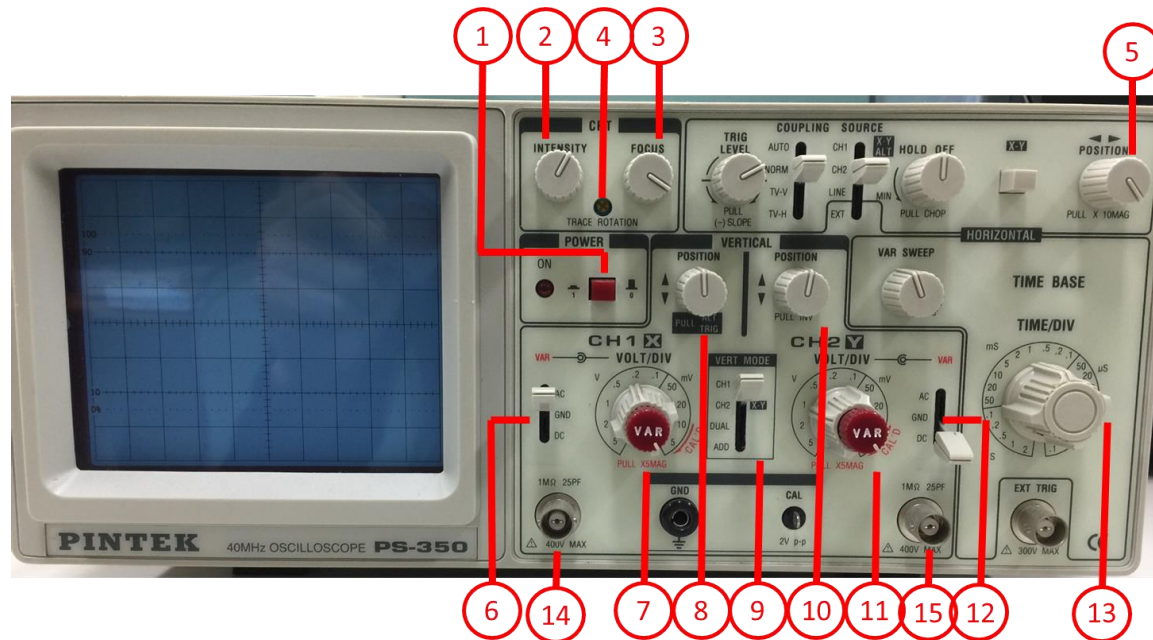
1. Power switch
2. Internal AM/FM adjustment knob
3. Synchronous TTL Pulses (Clock) Output
4. Ext. freq. input for freq. counter function
5. DC offset adjustment knob
6. External modulation input
7. Amplitude adjustment knob
8. External input BNC connector for Trigger/Gate and PSK/FSK (CMOS level)
9. Function Output BNC Connector (50 output impedance)
10. Rotary with Push Button

Function Generator (2/2) (Optional)



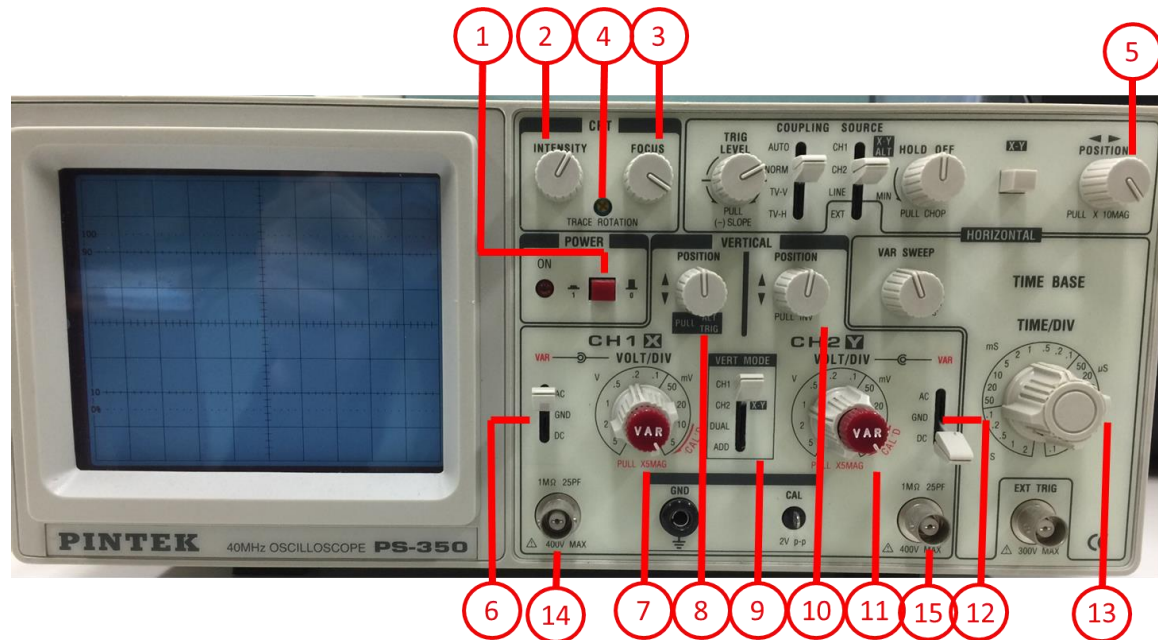
11. LCD Display
12. a. Change to the next selection.
b. In frequency editing, the frequency will $(/10)/(\times 10)$ if the cursor is off.
c. In frequency editing, the cursor goes to the left/right position if the cursor is on.
13. Select the output of sine, square, triangle or DC.
14. Select Amplitude / DC Offset
15. Enter the Sweep menu to select and set the linear or logarithm frequency sweep.
16. Trigger/Gate and PSK/FSK key
17. Fstep/Attn: Frequency Step/Attenuation Key
18. Counter/Sub Func: Auxiliary Function Key
19. Amplitude: Output Attenuation LED Indicator

Oscilloscope (1/3) (Optional)



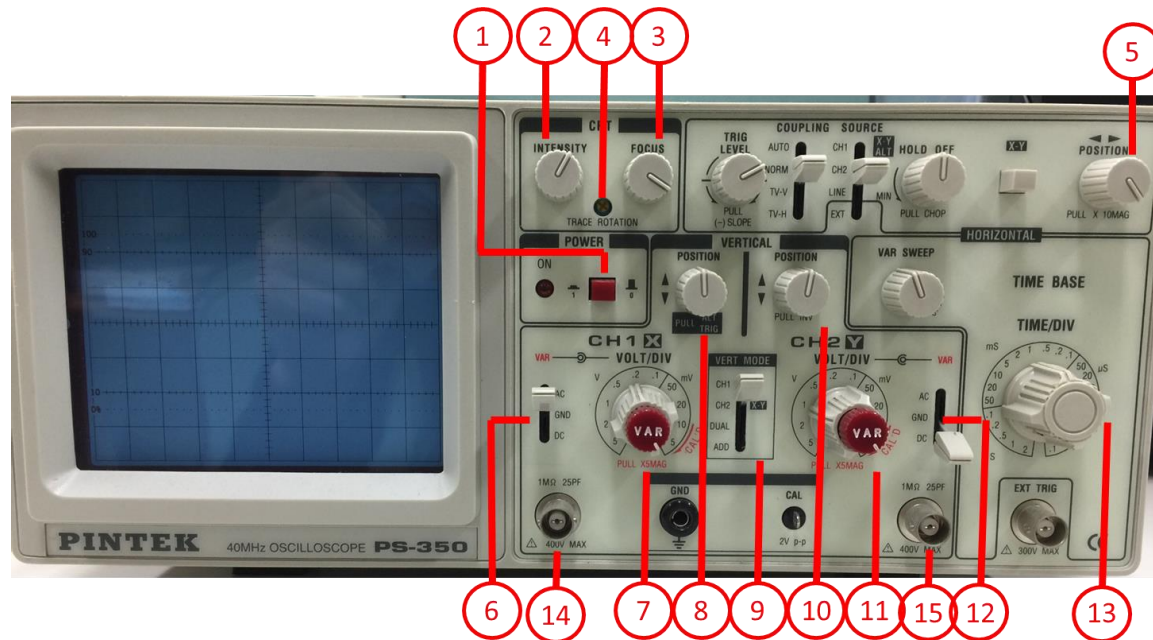
- 1 POWER Pushbutton :
The LED next to "ON" will light up, and the LED next to "OFF" will turn off.
- 2 INTENSITY : Horizontal Line Trace Adjustment (If the horizontal line is too thin, unclear, or invisible, adjust this knob.)
- 3 FOCUS : Focus Adjustment for the Horizontal Line (If the horizontal line is blurry, adjust this knob.)
- 4 TRACE ROTATION : Maintain Balance Between Horizontal and Vertical Axes (If the horizontal and vertical axes are misaligned, adjust this knob using a screwdriver.)
- 5 Horizontal Axis Left/Right Shift

Oscilloscope (2/3) (Optional)



- 6 12 Input Coupling Switch "DC-GND-AC"
(Select the input coupling mode for CH 1 and CH 2.)
- 7 11 Vertical Deflection Fine Adjustment "VERT VAR" :
Adjustable to 1/3 of the panel's indicated value,
allowing continuous adjustment.
For measurement, set it to the CAL position.
Pulling this knob activates the 5x magnification
function (X5 MAG).
- 8 10 Channel 1 & 2 Vertical Position Adjustment "CH 1, 2
POSITION" :
Adjusts the vertical position of the CH 1 and CH 2 scan
lines. Pulling up this knob inverts the CH 1 and CH 2
waveforms.

Oscilloscope (3/3) (Optional)



- 9 Vertical Operation Mode Selector Switch "VERT MODE" :
 CH 1: Single-trace scan for CH 1.
 CH 2: Trace scan for CH 2 (used with X-Y function operation).
 DUAL: Alternating (ALT) dual-trace scanning of CH 1 and CH 2.(When the HOLD OFF knob is pulled up, it switches to chopped (CHOP) dual-trace scanning.)
 ADD: Measures the summed signal of CH 1 and CH 2.
- 13 Time Base Fine Adjustment "TIME BASE VAR" :
 Provides continuous fine adjustment of the scan rate up to 5x magnification.
- 14 Channel 1 or X Input "CH1 or X INPUT" (14):
 BNC input connector for CH 1 or X-axis.
- 15 Channel 2 or Y Input "CH2 or Y INPUT" (15):
 BNC input connector for CH 2 or Y-axis.

Passive Components

- Electronic components that are with the properties
 - do not require an external power source
 - do not amplify or generate electrical signals
 - only dissipate, store, or modify energy
- Example
 - Capacitor (電容器)
 - Inductor (電感器)
 - Resistor (電阻器)

Capacitor

- Circuit Theory
 - In circuit theory, given a voltage, the ability of a capacitor to **store charge** (電荷) is called capacitance (電容), denoted as C .
- Unit
 - The unit of the capacitance is farad (法拉), denoted as F.
- Calculation
 - $C = Q / V$.
 - 1 farad is equal to 1 coulomb (庫倫) per volt, meaning a capacitor with a capacitance of 1 farad.



Types of Capacitors (Optional)

Type	Aluminum Electrolytic Capacitor (鋁質電解電容)	Ceramic Capacitor (陶瓷電容)	Tantalum Electrolytic Capacitor (鉭質電解電容)
Appearance		 $103 \rightarrow (10 \times 10^3) \text{ pF} = 0.01 \mu\text{F}$	 $227 \rightarrow (22 \times 10^7) \text{ pF} = 2.2 \mu\text{F}$
Dielectric Material	Aluminum Oxide	Various Ceramics	Tantalum Pentoxide (五氧化二鉭)
Operating Voltage	4 ~ 400V	6.3 ~ 250V	2.5 ~ 50V
Capacitance Range	47 ~ 10000 μF	0.001 ~ 100 μF	0.47 ~ 1000 μF
Advantages	•High tolerance, wide capacitance range •Low cost	•Excellent high-frequency characteristics •Non-polarized	•Small size with high capacitance •Stable capacitance
Disadvantages	•Leakage may cause performance issues •Large size •Polarized	•Capacitance varies significantly •Prone to cracking and defects	•Failure results in short circuit mode •Polarized

Inductors

● Circuit Theory

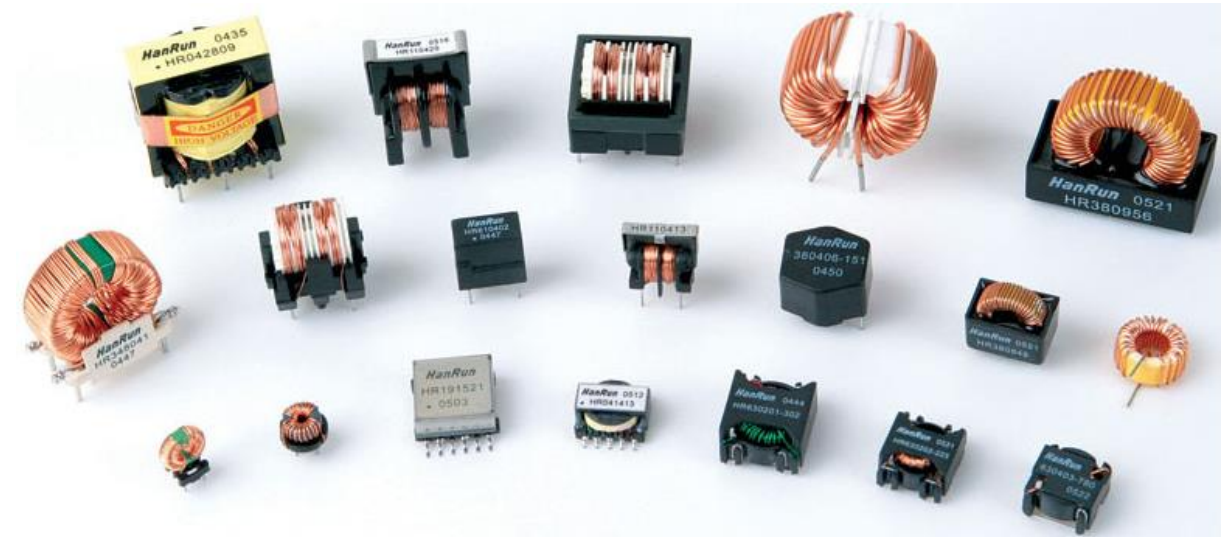
- An inductor, also known as an inductance coil (電感線圈), is the ability of a coil to **induce current** when it moves within a magnetic field.
- Its primary function is to isolate and filter AC signals and is denoted as L .

● Unit

- The unit of the inductance (電感) is henry, denoted as H.

● Calculation

- $v(t) = -L(di(t) / dt)$
 - $v(t)$ is the voltage at time t
 - L is the inductance
 - $i(t)$ is the current at time t

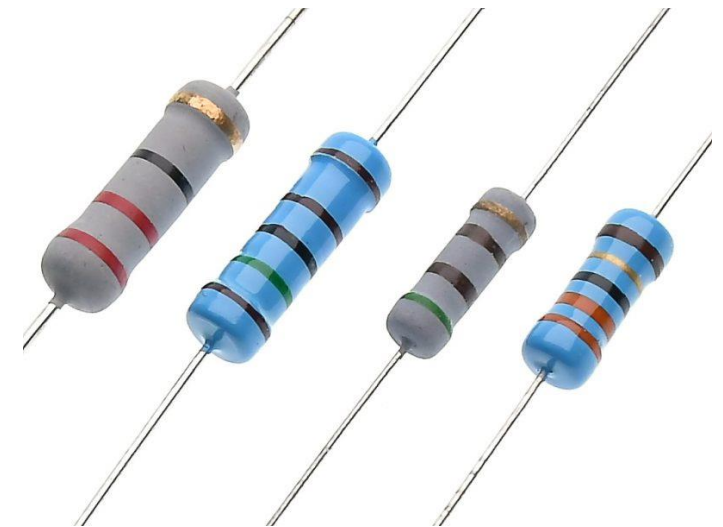


Classifications of Inductors (Optional)

- Classification by Structure
 - fixed inductors and tunable inductors.
 - wire-wound inductors (繞線電感) and non-wire-wound inductors (非繞線電感)
 - multilayer chip inductors (積層晶片電感) and printed inductors (印刷電感).
- Classification by Mounting Method
 - surface-mount inductors and through-hole (plug-in) inductors.
 - shielded inductors and non-shielded inductors.
- Classification by Operating Frequency
 - high-frequency inductors, mid-frequency inductors, and low-frequency inductors.
- Classification by Application
 - Oscillation inductors, rectification inductors, image deflection inductors, current-limiting inductors, resistor inductors, filter inductors, isolation inductors, compensation inductors, power inductors are commonly used.

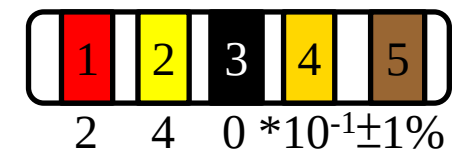
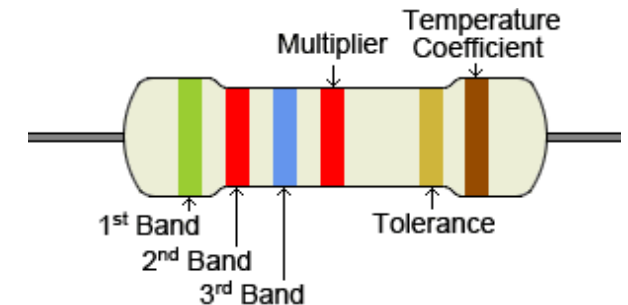
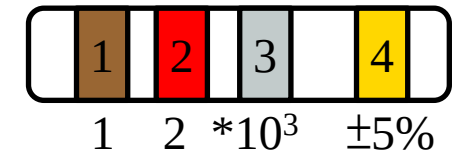
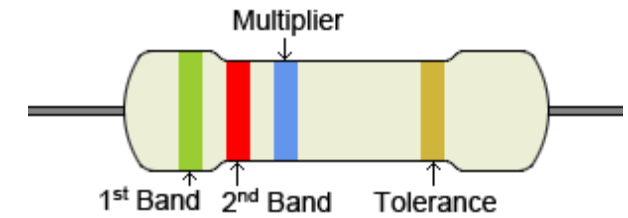
Resistor

- Circuit Theory
 - Resistor refers to any electronic or electrical component used to generate resistance (阻抗) and is denoted as R .
- Unit
 - The unit of resistance is ohm, denoted as Ω .
- Ohm's law (歐姆定律)
 - $R = V / I$
 - I is the current flowing through the conductor, measured in amperes (A).
 - V is the potential difference across the conductor, measured in volts (V).



Resistor Color Code

Color	1 st , 2 nd , and 3 rd Bands Significant figures	Multiplier	Tolerance		Temp. Coefficient (ppm/K)	
Black	0	$\times 10^0$	—		250	U
Brown	1	$\times 10^1$	$\pm 1\%$	F	100	S
Red	2	$\times 10^2$	$\pm 2\%$	G	50	R
Orange	3	$\times 10^3$	—		15	P
Yellow	4	$\times 10^4$	—		25	Q
Green	5	$\times 10^5$	$\pm 0.5\%$	D	20	Z
Blue	6	$\times 10^6$	$\pm 0.25\%$	C	10	Z
Violet	7	$\times 10^7$	$\pm 0.1\%$	B	5	M
Gray	8	$\times 10^8$	$\pm 0.05\%$	A	1	K
White	9	$\times 10^9$	—		—	
Gold	—	$\times 10^{-1}$	$\pm 5\%$	J	—	
Silver	—	$\times 10^{-2}$	$\pm 10\%$	K	—	
None	—	—	$\pm 20\%$	M	—	



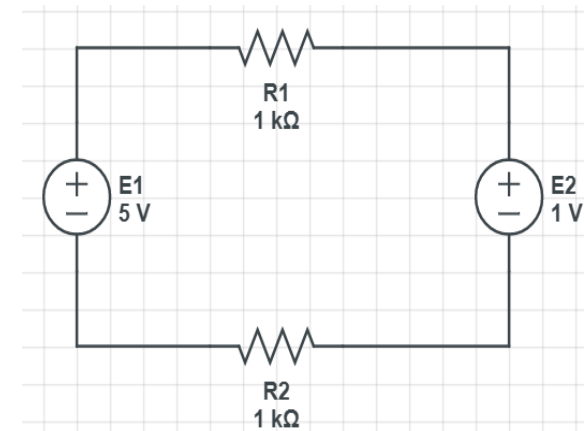
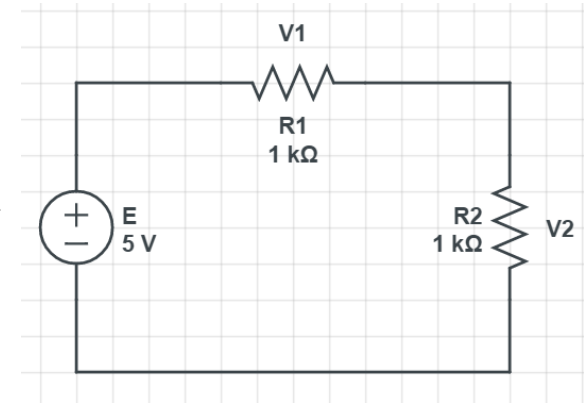
Series Resistors

- Collection of Theorems and Laws

- The current flowing through each component is equal.
- Kirchhoff's Voltage Law (KVL): In a closed loop, the sum of the source voltages equals to the total voltage drops across all components.
- The voltage drop across a component is proportional to its resistance value.

- Collection of formulas

- Total Resistance in series $\rightarrow R_T = R_1 + R_2 + R_3 + \dots$
- Current in a series circuit $\rightarrow I = I_1 = I_2 = I_3 = \dots$
- Total Voltage in a series circuit $\rightarrow E = V_1 + V_2 + V_3 + \dots$
- Total Power in a series circuit $\rightarrow P_T = P_1 + P_2 + P_3 + \dots$
- Current in a series circuit with Multiple Voltage Sources $\rightarrow I = \frac{E_1 + E_2 + \dots}{R_1 + R_2 + \dots}$
- Voltage Division Formula $\rightarrow V_1 = \frac{E}{R_1 + R_2} \times R_1$ 、 $V_2 = \frac{E}{R_1 + R_2} \times R_2$



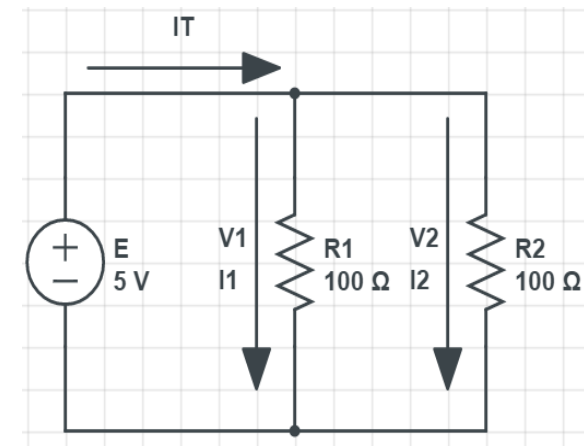
Parallel Resistors

• Definition

- In a circuit, when two or more components have one end connected to a common point and the other end also connected to another common point, this configuration is called parallel connection.

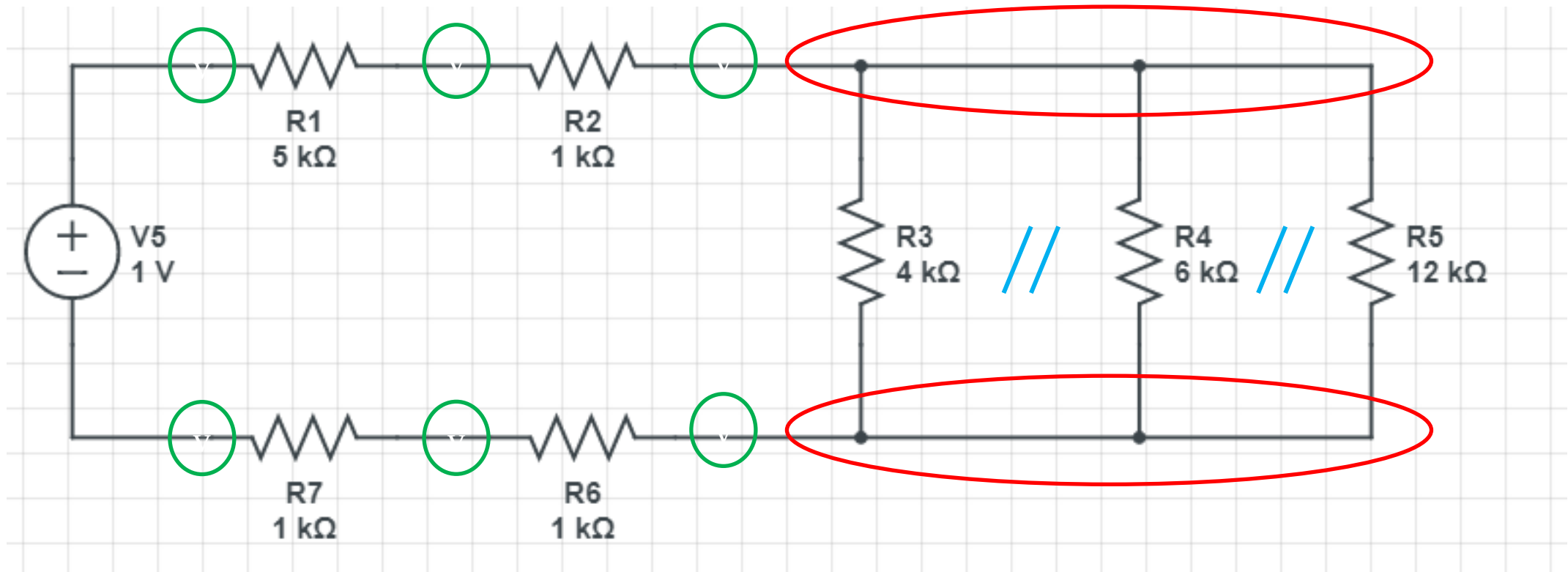
• Characteristics

- The **same voltage drop** across each component in a parallel circuit, i.e., $V_1 = V_2 = E$
- Each component in a parallel circuit has its own **individual current**, which may not necessarily be equal. In figure, the currents through resistors R_1 and R_2 are different.
- The total current is the sum of all branch currents, i.e., $I_T = I_1 + I_2$, where $I_1 = \frac{E}{R_1}$ and $I_2 = \frac{E}{R_2}$.
- The reciprocal of the total resistance is equal to the sum of the reciprocals of each branch resistance, i.e., $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$ or $R_T = \frac{R_1 R_2}{R_1 + R_2}$.



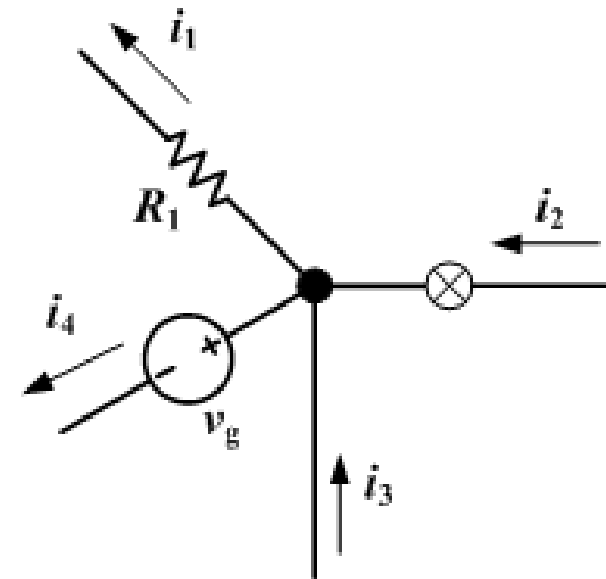
Series and Parallel Resistor Calculations

- Calculate the total resistance (R_T), total current (I_T), voltage (V) across each resistor, and current (I) through each resistor



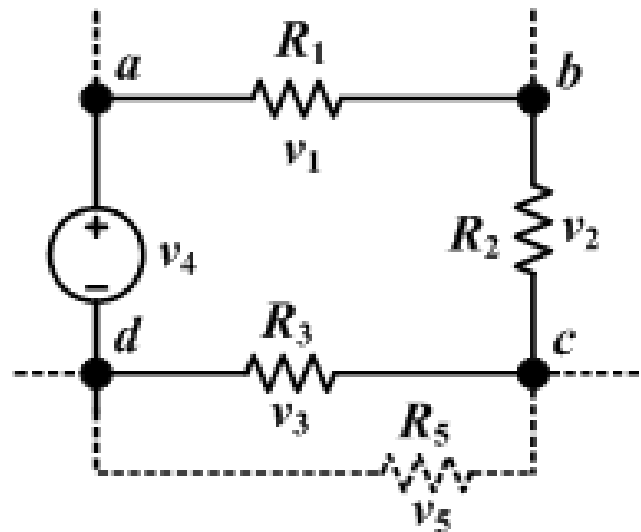
Kirchhoff's Current Law (KCL)

- Also known as Kirchhoff's First Law
- The sum of all currents entering a node equals the sum of all currents leaving the node.
- In the example below, the total current leaving the black dot equals the total current entering it, i.e., $i_1 + i_4 = i_2 + i_3$



Kirchhoff's Voltage Law (KVL)

- Also known as Kirchhoff's Second Law.
- The algebraic sum of the voltage differences (voltages) across all components in a closed loop equals zero.
- In the given example, the sum of the voltages across all components in the closed loop equals zero, i.e., $v_1 + v_2 + v_3 - v_4 = 0$

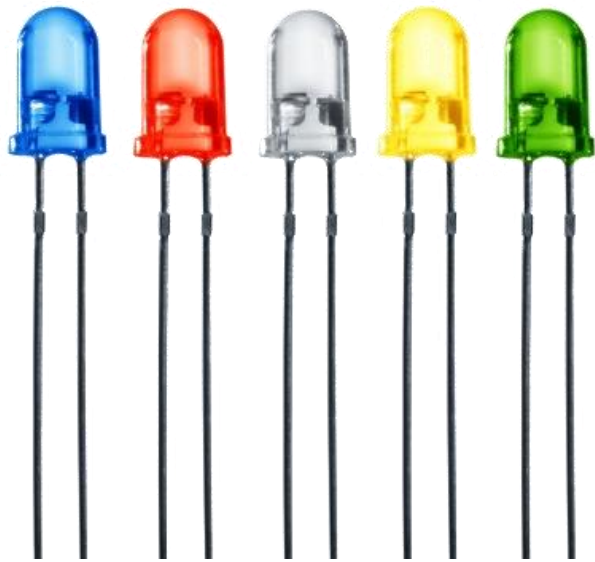


Active Components

- Electronic components that are with the properties
 - require an external power source to function
 - are capable of controlling the flow of signals
 - are capable of amplifying signals.
- Example
 - Light-emitting diodes (LEDs)
 - Transistors (電晶體)
 - Integrated circuits (ICs)
 - Logic gates
 - Operational amplifiers (OP Amps)

Light-Emitting Diode (1/3)

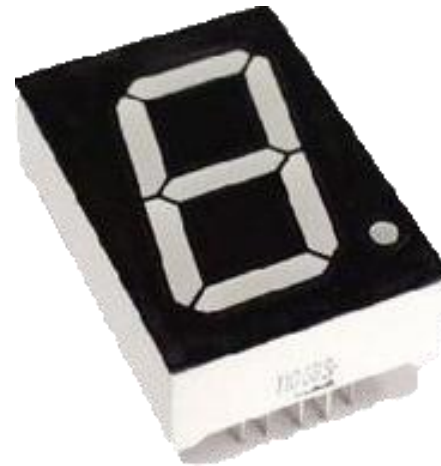
- Classification: monochrome LED, bicolor LED, seven-segment display, dot matrix LED
- How to Use: **Measurement** (Use a multimeter to test functionality)



Monochrome LED



Bicolor LED



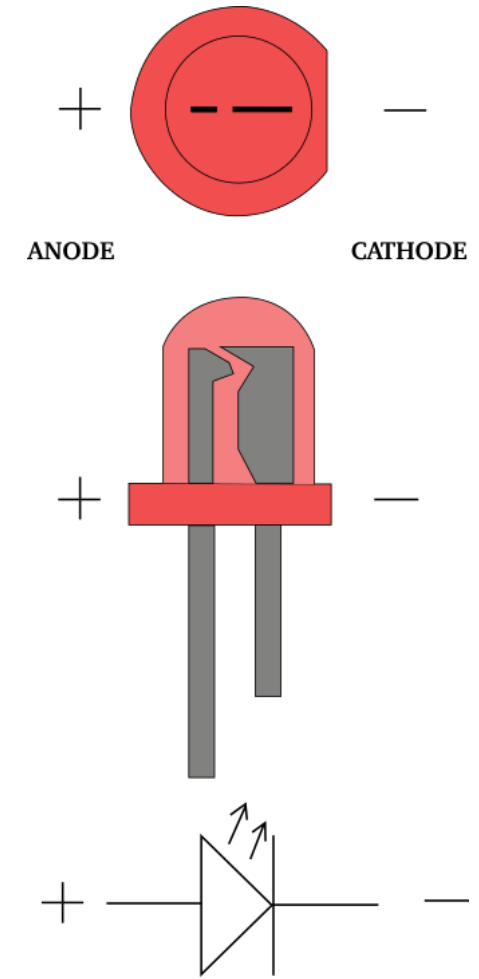
Seven-Segment Display



Dot Matrix LED

Light-Emitting Diode (2/3)

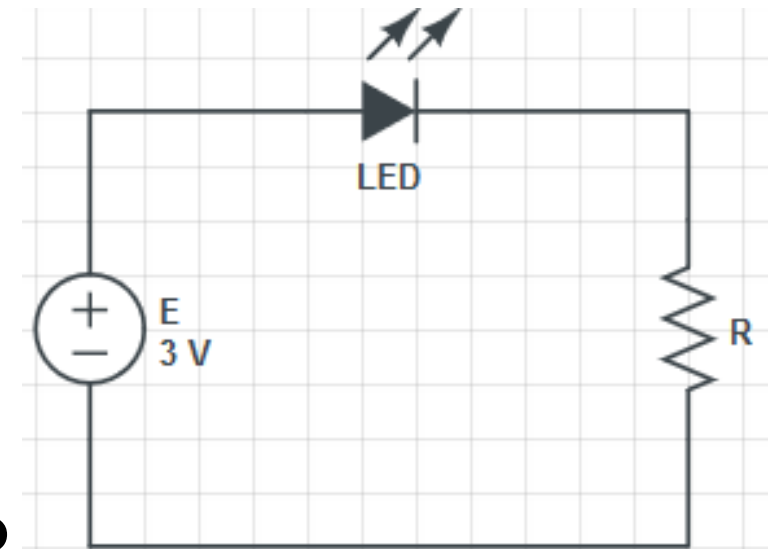
- Remember to connect a resistor with the LED.
- LED conducts electricity in only one direction, which is called forward bias.
- The operating voltage of an LED is approximately **1.6 V**, and the operating current is between 25-30 mA (using a red LED as an example).



Light-Emitting Diode (3/3)

Resistor Value Calculation

- How to calculate the resistor value?
- Since the operating voltage of an LED is approximately 1.6V and the operating current is around 25-30 mA, according to the KVL:
 - If 25 mA is designed, $R = (3V - 1.6V)/25 \text{ mA} = 56 \Omega$
 - If 30 mA is designed, $R = (3V - 1.6V)/30 \text{ mA} \cong 47 \Omega$
 - Thus, the resistor value R can be chosen between 40Ω and 33Ω .

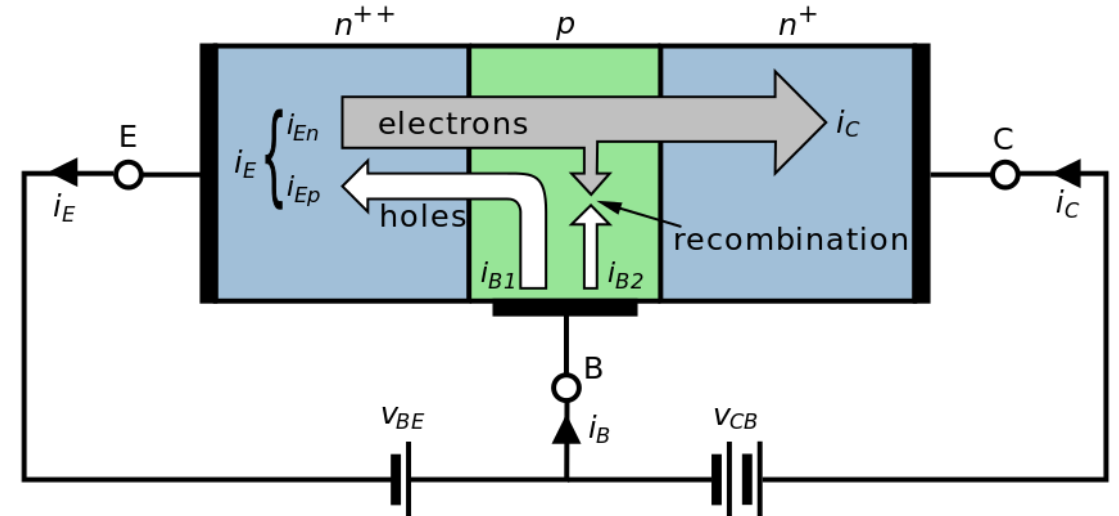


Transistor (Optional)

- Applications: amplifier or switch.
- Types of transistors:
 - Power transistors
 - Low-frequency power transistors: Mainly used for audio amplification, such as in radios and stereo amplifiers.
 - High-frequency power transistors: Mainly used in RF communication transmission circuits, such as oscillators and frequency multipliers.
 - Small-signal transistors
 - Low-frequency small-signal transistors: Mainly used for low-frequency amplification in audio equipment, medical instruments, and physical/chemical instruments.
 - High-frequency small-signal transistor: Typically used for amplification and oscillation in applications above 100 KHz.

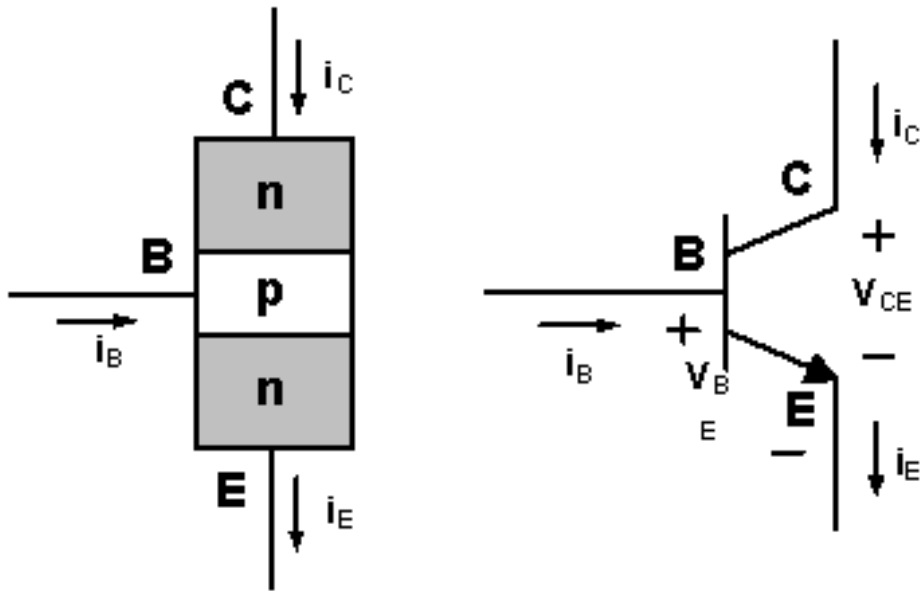
Transistor (BJT) (1/3) (Optional)

- A Bipolar Junction Transistor (BJT) commonly known as a transistor.
 - Controlled by current
 - Three terminals: emitter (E), base (B), and collector (C).
- The gray arrow indicates **the direction of electron flow**, which is from the emitter to the collector, with a small number recombining with holes in the base.
- The white arrow shows **the direction of hole flow** (the majority carriers) in the base.
- For an **NPN transistor** to be in the active region, the base-emitter junction must be forward-biased (positive voltage on the base relative to the emitter), and the collector-base junction must be reverse-biased (negative voltage on the base relative to the collector).

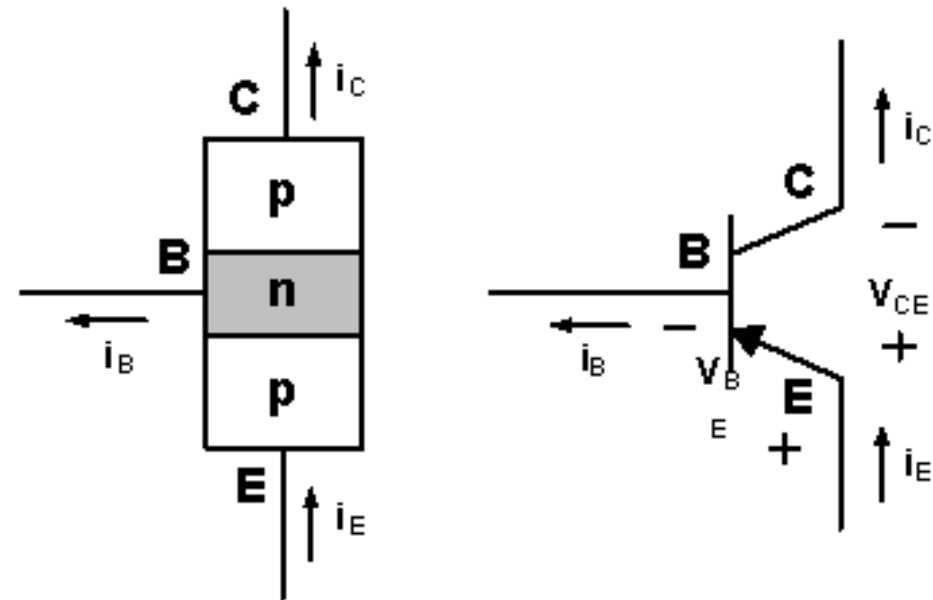


Transistor (BJT) (2/3) (Optional)

- BJTs have two types: **NPN** and **PNP**.
 - NPN Type: Consists of **two N-type regions** separated by a P-type region.
 - PNP Type: Consists of **two P-type regions** separated by an N-type region.



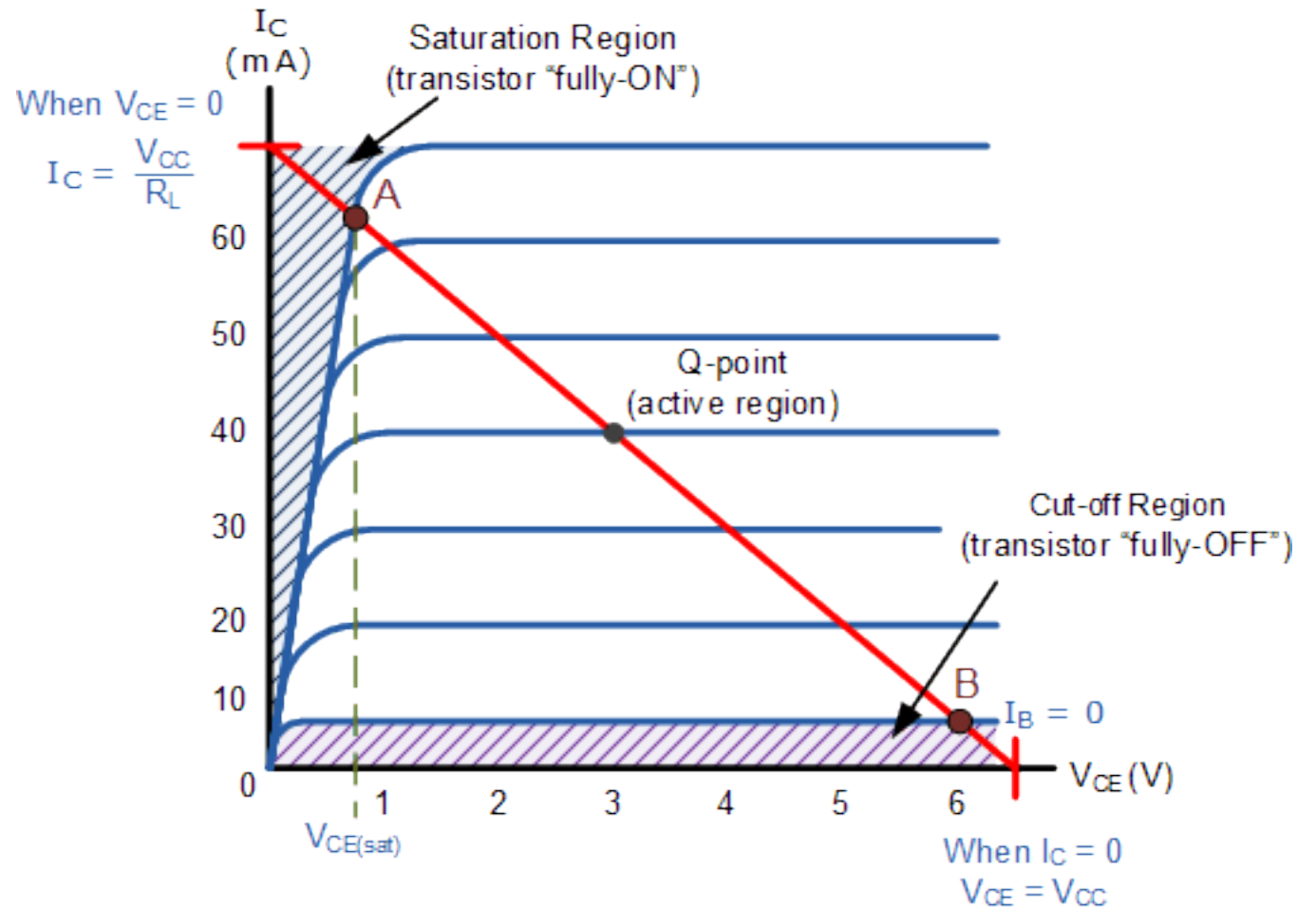
Circuit Symbol of NPN BJT



Circuit Symbol of PNP BJT

Transistor (BJT) (3/3) (Optional)

- A BJT operates in three different modes
 - Cut-off mode
 - Linear amplification mode
 - Saturation mode



Field Effect Transistor (FET) (1/2) (Optional)

● Properties

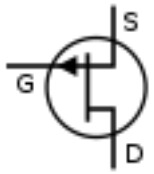
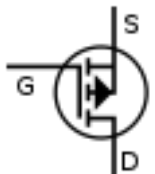
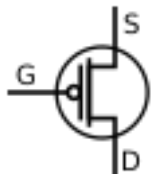
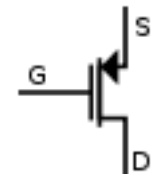
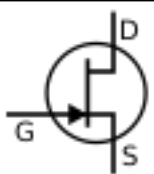
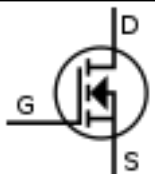
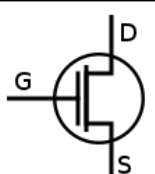
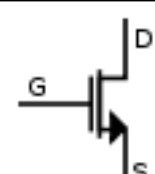
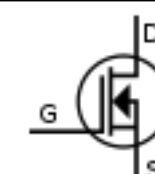
- Three terminals: Source (S), Drain (D), and Gate (G).
- FET controls electrical current by using an **electric field**.
- FETs are voltage-controlled devices, meaning they require very little input current at the gate terminal to function.

● Types of FET

- Junction Field-Effect Transistor (JFET)
 - Uses a PN junction to control current flow.
 - It can be N-channel (majority carriers: electrons) or P-channel (majority carriers: holes).
 - Normally ON (conducts when no gate voltage is applied). Requires a reverse bias voltage at the gate to turn OFF.
- Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET)
 - Has an insulating layer (silicon dioxide, SiO_2) between the gate and channel.
 - Available in Enhancement mode (normally OFF) and Depletion mode (normally ON).

Field Effect Transistor (FET) (2/2) (Optional)

- Circuit Symbol

					P-Channel
					N-Channel
JFET	Enhancement Mode MOSFET	Enhancement Mode MOSFET	Depletion Mode MOSFET	Depletion Mode MOSFET	

Integrated Circuits (ICs) (1/2) (Optional)

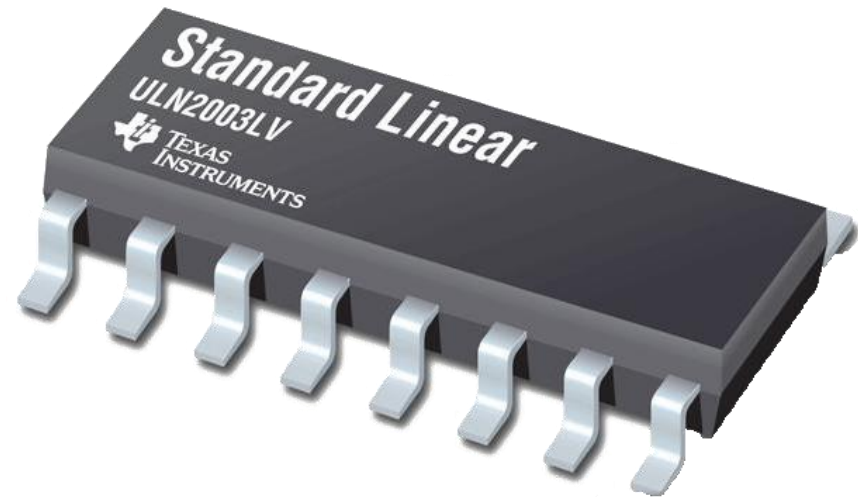
- An integrated circuit (IC) is the technology that integrates electronic components such as transistors, diodes, resistors, and capacitors onto a single chip by using microelectronics techniques.
- Characteristics of ICs
 - Small size, Powerful functionality, High reliability, Low cost, and Convenient to use
- Types of ICs
 - Consumer ICs: These focus on **versatility** and **innovation**.
 - Industrial ICs: These focus not only on **functionality** but also on **stability** and **standardization**, ensuring they are interchangeable when used in large quantities.

Integrated Circuits (ICs) (1/2) (Optional)

- IC Diagram



CPU

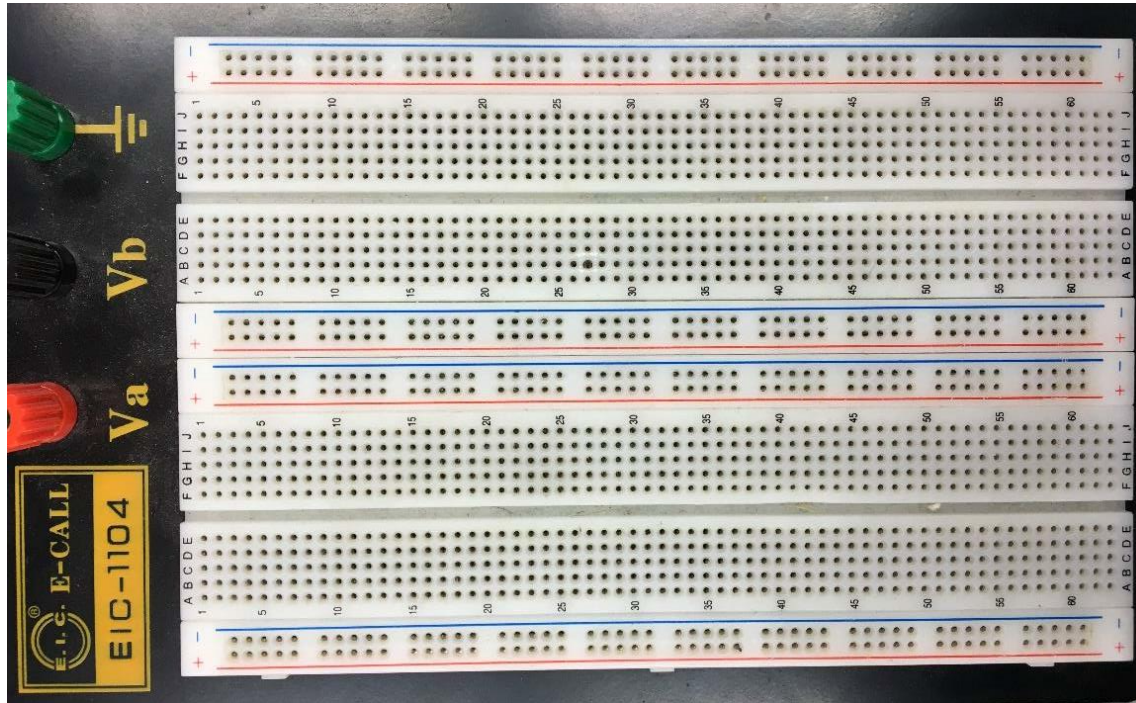


IC

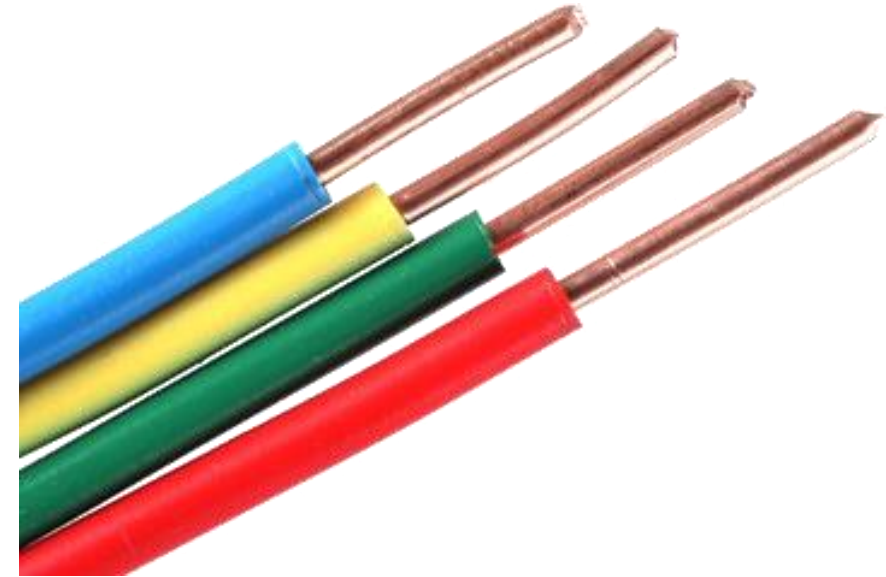
Breadboard (1/5)

Principle

- Breadboard Diagram



Breadboard

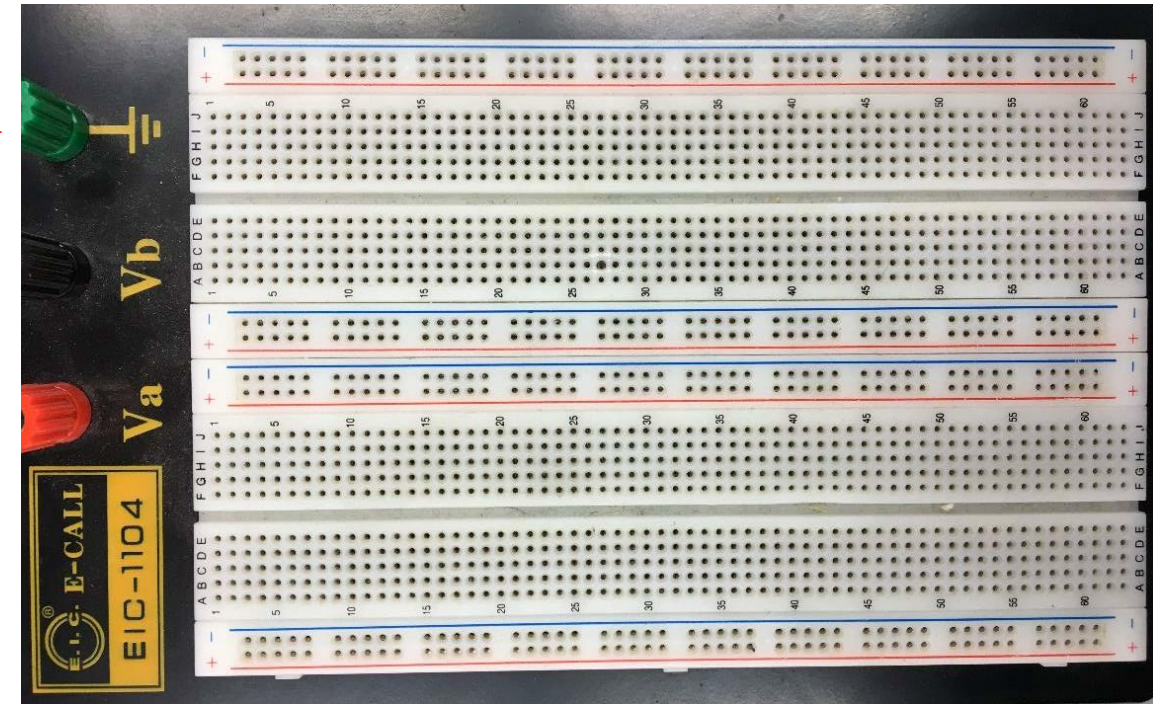


Single-core Wire

Breadboard (2/5)

Structure

- The entire board is made from thermosetting phenolic resin (熱固性酚醛樹脂), with **metal strips** on the underside.
- Holes are drilled at corresponding positions on the board so that when components are inserted, they make **contact with the metal strips, allowing for electrical conduction.** Typically, every five holes are connected by a metal strip.
- In the center of the board, there is usually a groove (凹槽), designed specifically for testing ICs or chips.
- On both sides of the board, there are two rows of sockets (power rails 電源軌), also grouped in sets of five, which provide power to the components on the board.

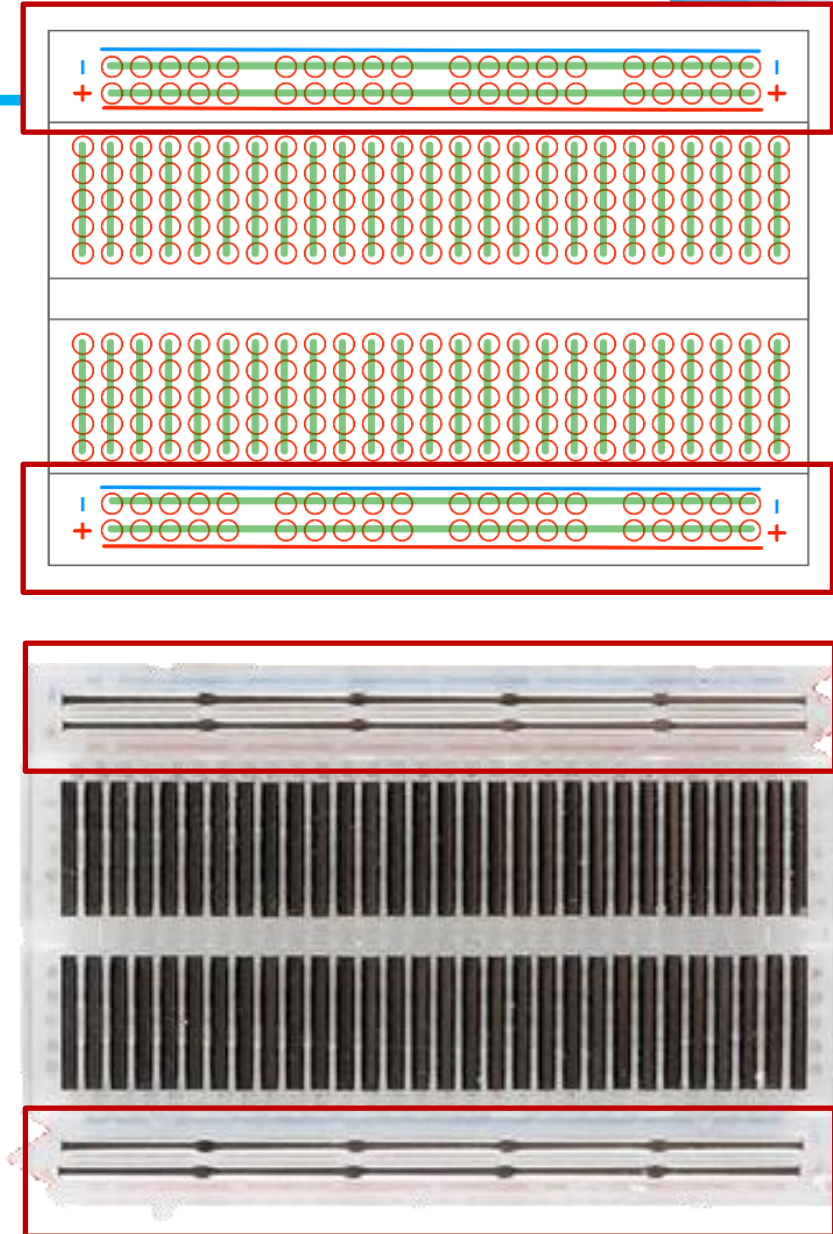


metal strips

Breadboard (3/5)

Power Rails (電源軌)

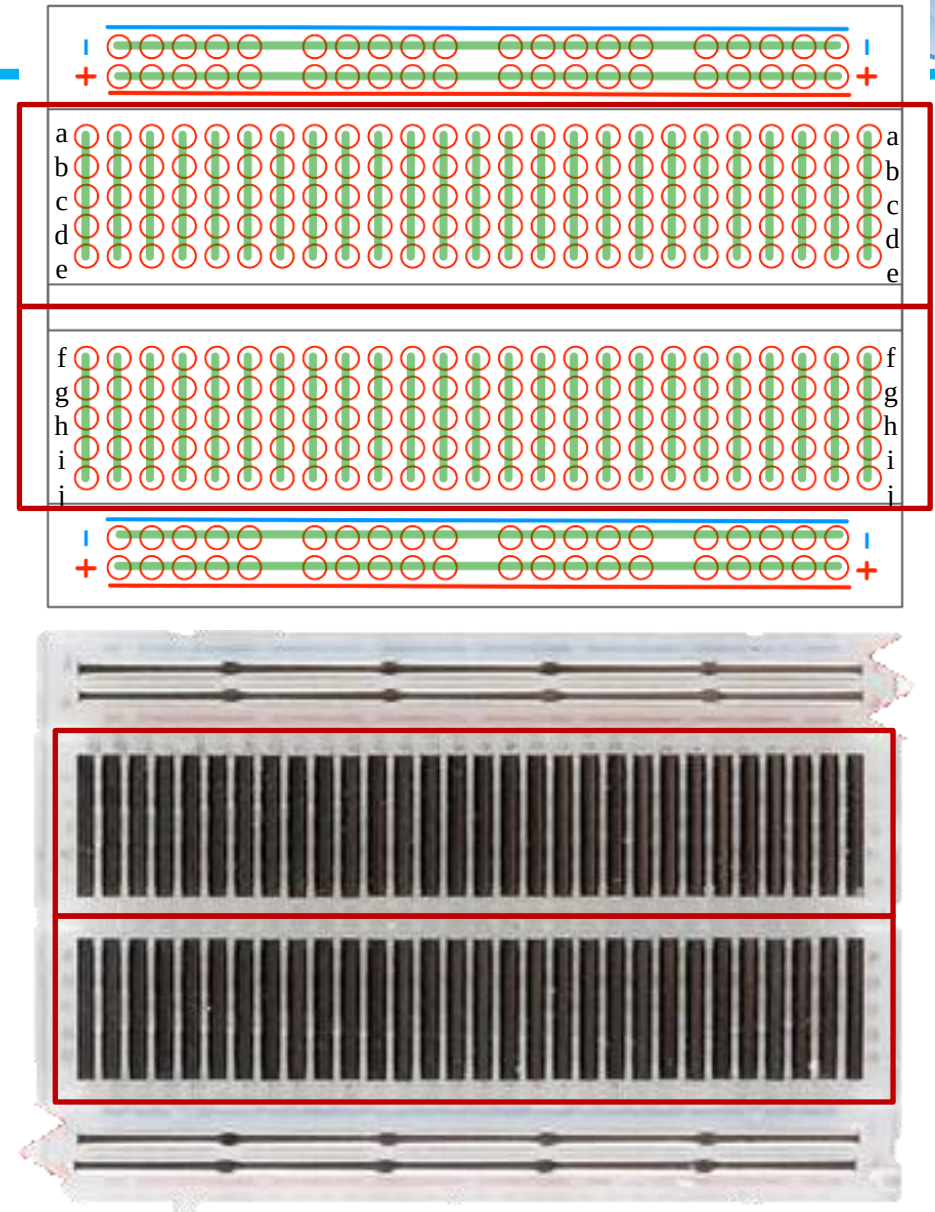
- On both the top and bottom sides of the breadboard, there are two rows of sockets, generally used for the power supply connections.
- Each row has 5 groups of sockets, each with 5 holes (internally connected)
 - Holes in the row marked with a “+” are all for the positive voltage.
 - Holes in the row marked with a “-”, are all for ground.
- When using the entire breadboard, it is common to connect the “+” rails together with wires and the “-” rails together with wires.



Breadboard (4/5)

Connection Rails (接線軌)

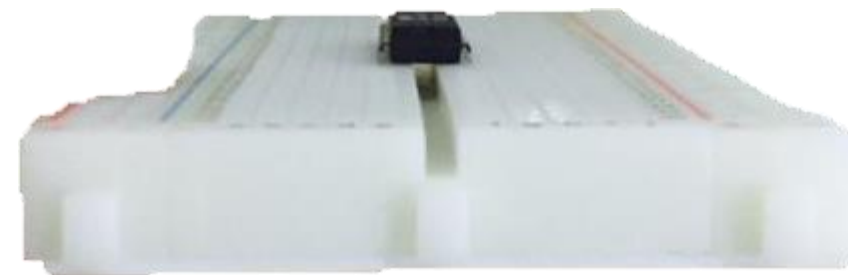
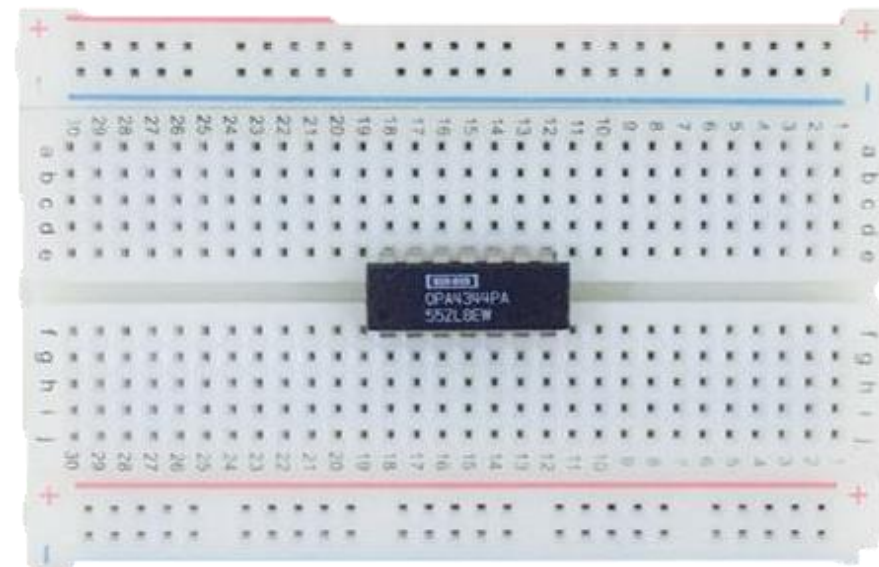
- The connection holes are divided into two parts: the upper and lower sections.
- They serve as the main working area for inserting components and jumper wires.
- The 5 sockets in the same column (i.e., a-b-c-d-e, f-g-h-i-j) are interconnected.
- The rows in the upper and lower sections of the groove (i.e., e-f) are not interconnected.



Breadboard (5/5)

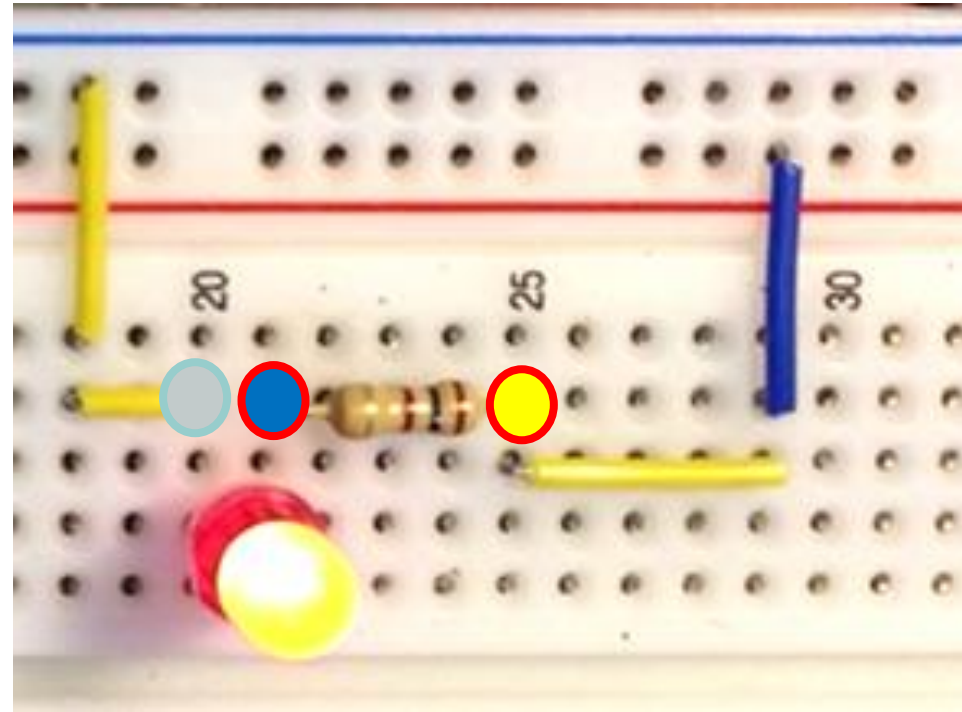
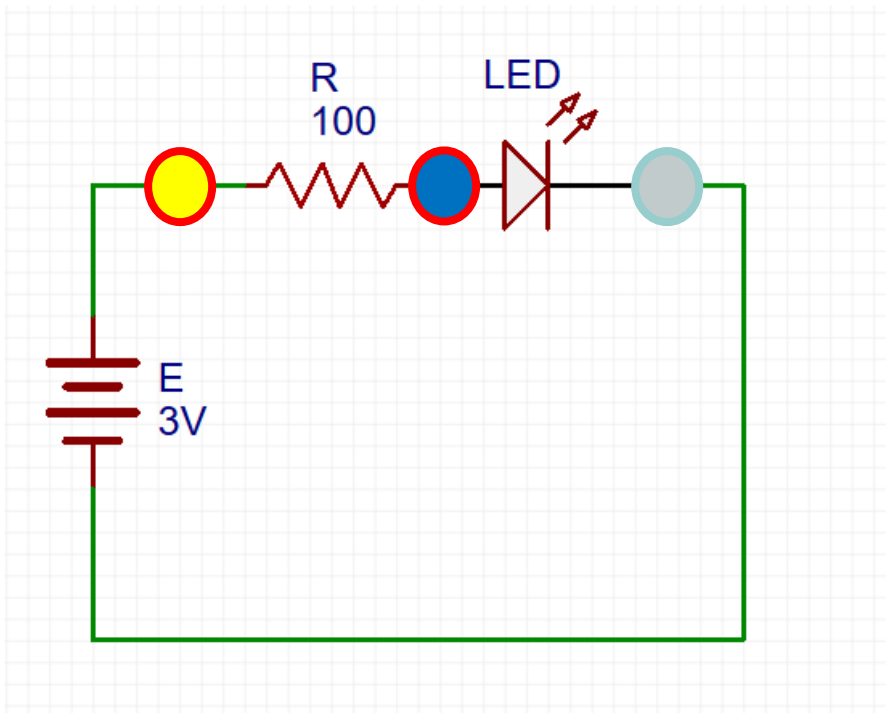
Groove (凹槽)

- Groove is in the middle of the breadboard.
- It indicates that the upper and lower sections are disconnected.
- It ensures that the gap between the two sections is 7.26mm for inserting a standard narrow-body DIP (Dual Inline Package) IC.
- It allows tools like tweezers to be inserted underneath the IC, making it easier to remove the IC without damaging it.



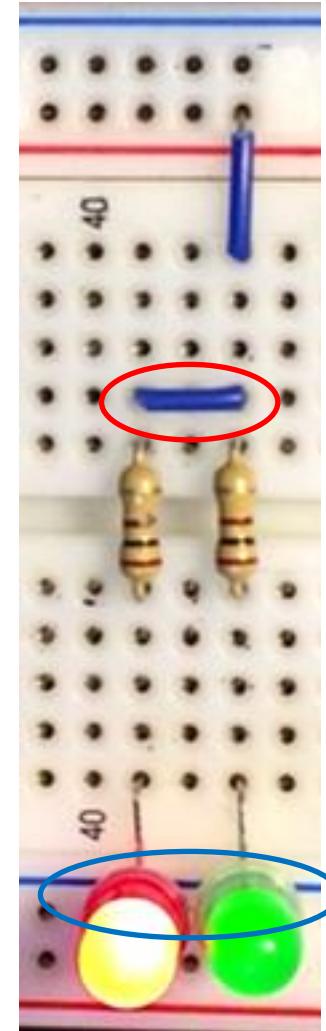
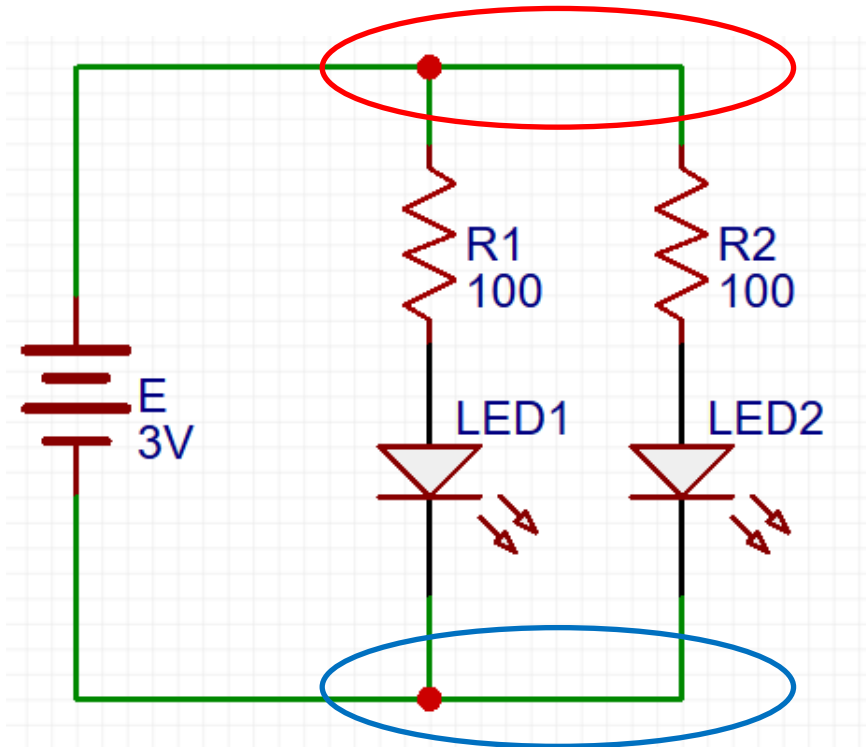
Wiring Example (1/3)

Series Circuit



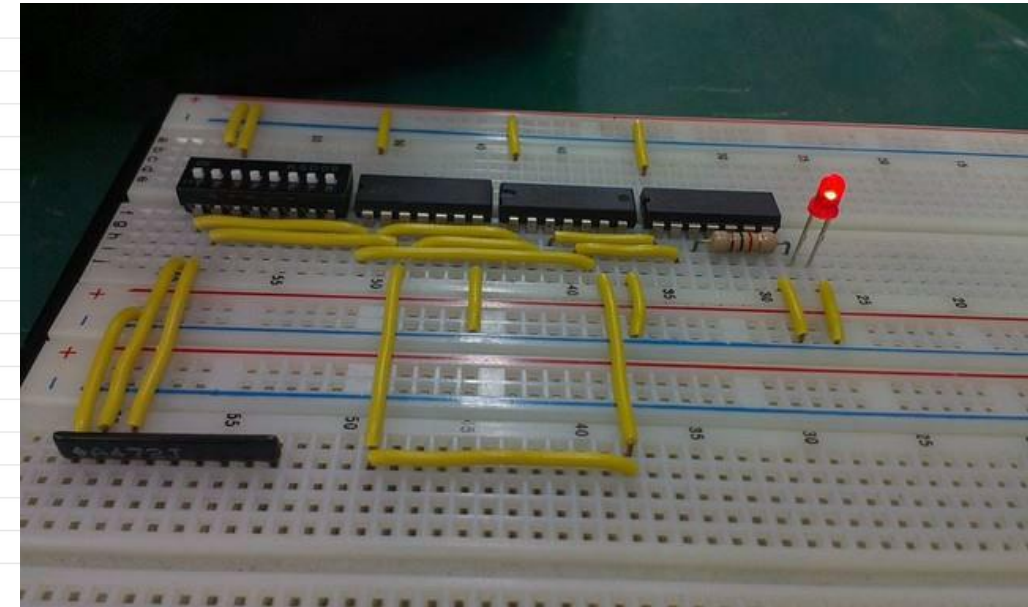
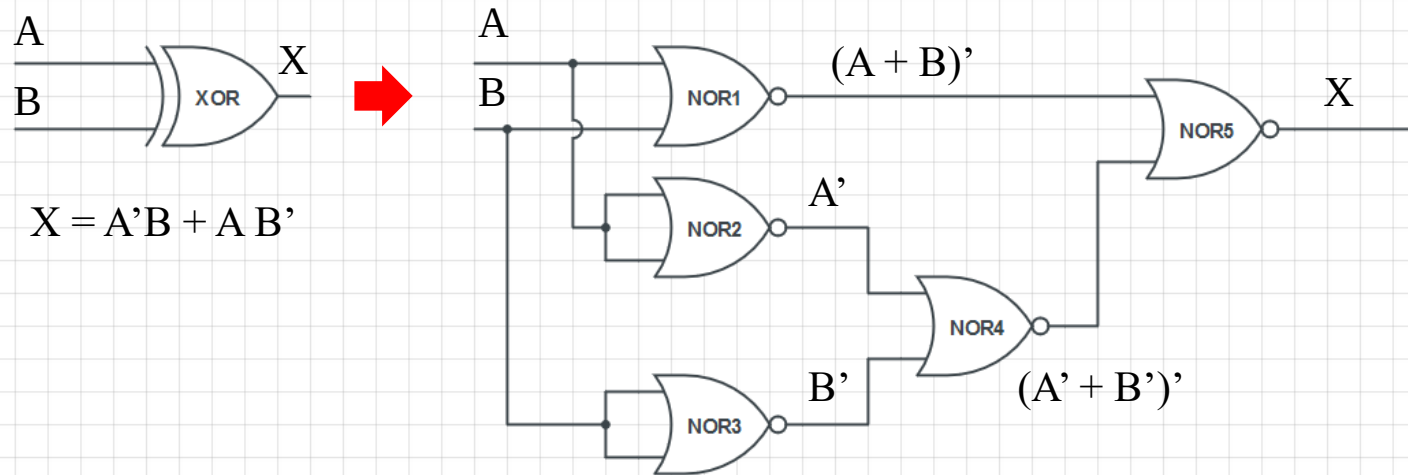
Wiring Example (2/3)

Parallel Circuit



Wiring Example (3/3)

Implement XOR Using NOR

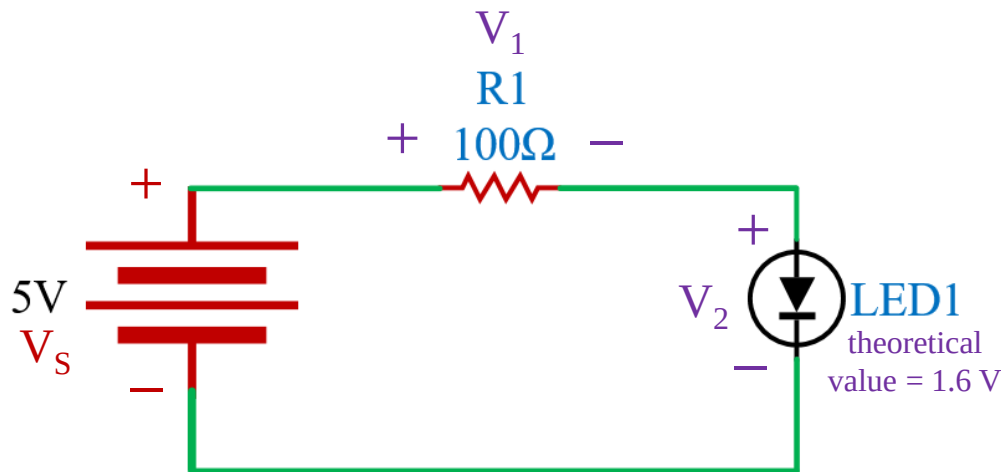


$$\begin{aligned} X &= ((A + B)' + (A' + B'))' \\ &= (A + B)(A' + B') \\ &= AA' + AB' + BA' + BB' \\ &= AB' + A'B \end{aligned}$$

=

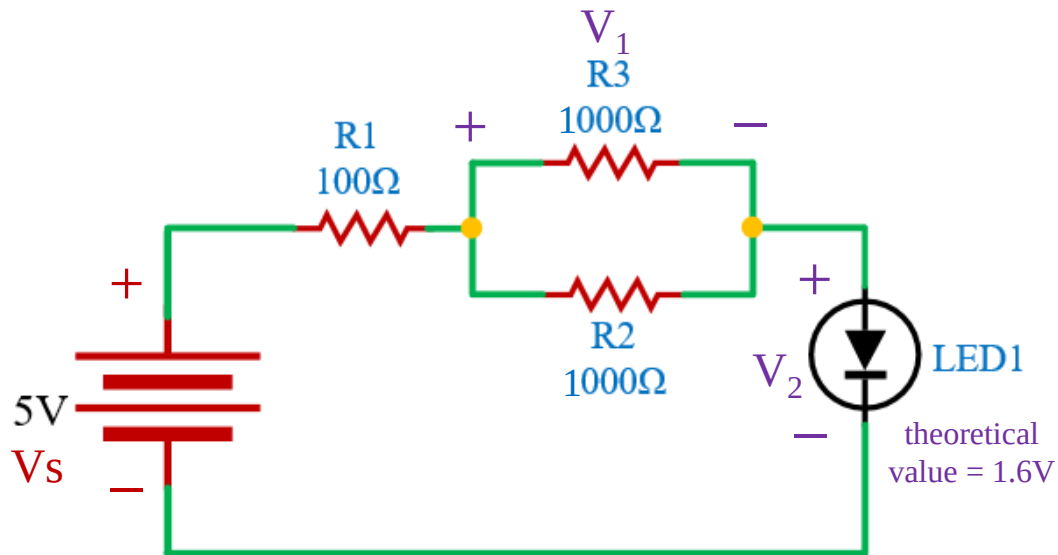
$$\begin{aligned} X &= \overline{\overline{A + B}} + \overline{\overline{A' + B'}} \\ &= (A + B) \cdot (\overline{A} + \overline{B}) \\ &= A\overline{A} + A\overline{B} + B\overline{A} + B\overline{B} \\ &= A\overline{B} + A\overline{B} \end{aligned}$$

LAB0-1



- Connect the LED1 in series with a $100\ \Omega$ resistor (R1). The LED1 will emit light.
- Use the multimeter to measure the following data:
 - V_1 is the voltage across R1.
 - V_2 is the voltage across LED1.
 - V_s is the total voltage.
- Calculate the theoretical values for **current**, **voltage**, and **resistance**, and then compare them with the measured results.
- Write the results on the calculation sheet and take a photo.

LAB0-2



- Connect the LED1, a 100Ω resistor ($R1$), and two parallel 1000Ω resistors ($R2$, $R3$) in series. The LED1 will emit light
- Use the multimeter to measure the following data:
 - V_1 is the voltage between the two yellow dots.
 - V_2 is the voltage across LED1.
 - V_s is the total voltage.
- Calculate the theoretical values for the **current**, **voltage**, and **resistance**, and then compare them with the measured results.
- Write the results on the calculation sheet and take a photo.

Homework Submission Guidelines

- Please **download the lab report template from TronClass** and follow the format specified in the template.
- Before submitting your lab report, **convert it into a PDF file** and then upload it.
- **The deadline for uploading lab report is from the end of the current class to before the next class.** Late submissions will not be accepted.
- File names **should follow the format: LAB#_StudentID**, e.g., LAB1_00857001.
- If the assignment requires images, **ensure the images are complete and clear** (e.g., circuit connections are clear, and the LED is visibly light).